



NORTHERN BEACHES SECONDARY COLLEGE

MANLY SELECTIVE CAMPUS

HIGHER SCHOOL CERTIFICATE

Trial Examination

2018

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3hours
- Use **black** pen
- Write your Student Number at the top of each page
- Section I – Multiple Choice – use the Answer Sheet provided
- Section II – Free Response – use a separate booklet for **each** question.
- NESA approved calculators and templates may be used.
- Reference sheet provided.

Section I Multiple Choice

- 10 marks
- Attempt all questions

Section II – Free Response

- Questions 11-16 – 90 marks
- Each question is of equal value
- All necessary working should be shown in every question.

Weighting: 40%

Section 1 Multiple Choice: Attempt Questions 1 – 10

Answer questions on the provided answer sheet.

Allow approximately 15 minutes for this section.

- Q1 The point M is the midpoint of the points A and B . The coordinates of A are $(2, -3)$ and the coordinates of M are $(-2, 1)$. The coordinates of B are:

A $(-6, 5)$

B $(5, -6)$

C $(0, -1)$

D $(-1, 0)$

- Q2 The correct solution to $|2x + 3| \geq 7$ is:

A $-2 \leq -x \leq 5$

B $x \leq -2, x \geq 5$

C $x \leq -5, x \geq 2$

D $-2 \leq x \leq 2$

- Q3 The coordinates of the focus of the parabola $4y = (x + 2)^2 - 4$ is:

A $(-2, 0)$

B $(-2, -1)$

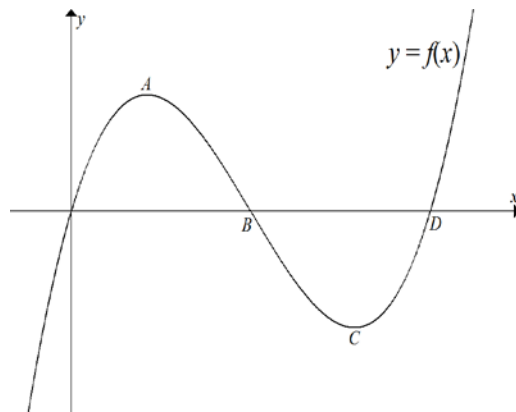
C $(-2, -3)$

D $(2, -4)$

Q4 If $\int_0^a (4 - 2x) \, dx = 4$, find the value of a .

- A $a = -2$
- B $a = 0$
- C $a = 4$
- D $a = 2$

Q5 The diagram shows the graph of the function $y = f(x)$.



At which point is $f'(x) > 0$ and $f''(x) > 0$?

- A A
- B B
- C C
- D D

- Q6 When the curve $y = e^x$ is rotated about the x – axis between $x = -2$ and $x = 2$, the volume of the solid generated is given by:

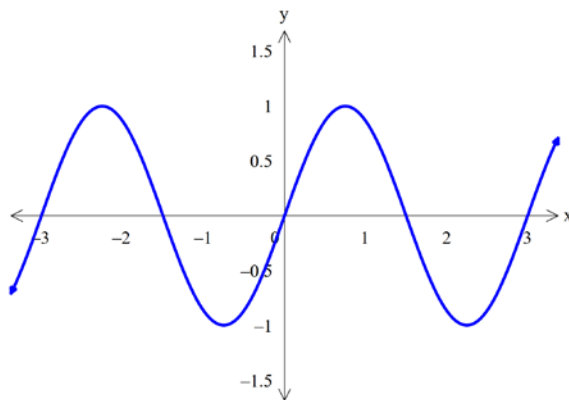
A $\pi \int_{-2}^2 e^x dx$

B $2\pi \int_0^2 e^{x^2} dx$

C $\pi \int_{-2}^2 e^{x^2} dx$

D $\pi \int_{-2}^2 e^{2x} dx$

- Q7 In the diagram, the graph of the function $y = \sin(kx)$ is given.



Which could be the value of k ?

A $\frac{3\pi}{2}$

B $\frac{2\pi}{3}$

C 3

D $\frac{2}{3}$

Q8 Which expression is a term of the geometric series $2x - 4x^3 + 8x^5 - \dots$?

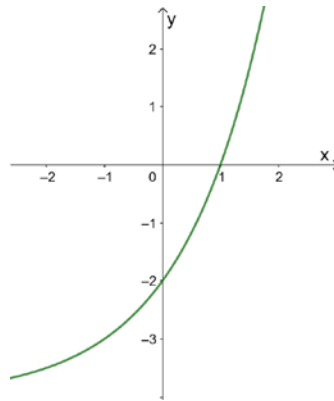
A $-2^{10} x^{19}$

B $2^{10} x^{19}$

C $-2^9 x^{19}$

D $2^9 x^{19}$

Q9 In the diagram, the graph of the function $y = 2^{x+a} + b$ is given



Which could be the values of a and b?

A $a = 1$ and $b = 2$

B $a = 1$ and $b = -4$

C $a = 1$ and $b = -2$

D $a = -1$ and $b = 4$

- Q10 The region bounded by the x -axis and the part of the graph $y = \cos x$ between $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$ is separated into two regions by the line $x = k$. If the area of the region for $-\frac{\pi}{2} \leq x \leq k$ is three times the area of the region for $k \leq x \leq \frac{\pi}{2}$ then the value of k equals ?

A $\sin^{-1} \frac{1}{4}$

B $\frac{\pi}{3}$

C $\frac{\pi}{4}$

D $\frac{\pi}{6}$

End of Multiple Choice

Section II**90 marks****Attempt Questions 11–16****Allow about 2 hours and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations

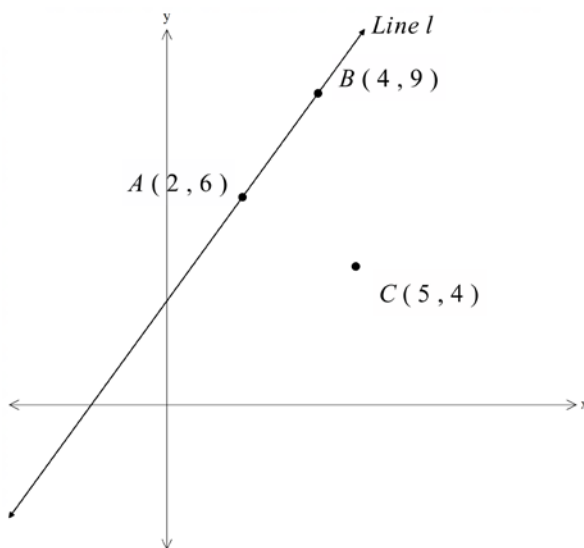
Question 11: Start A New Booklet**15 Marks**

- a. Factorise $12x^2 - 4x - 1$ 1
- b. Express $\frac{18}{3 - \sqrt{3}}$ in the form $a + \sqrt{b}$ where a and b are integers. 2
- c. Differentiate $\frac{2x + 1}{\sqrt{x}}$. 2
- d. Determine $\int_0^2 (5x - 1)^9 dx$ 3
- e. Determine the domain of the function $f(x) = \frac{x}{\sqrt{x^2 - 4}}$. 2
- f. Given $\cos(2\theta) = \frac{1}{2}$ for $0 \leq \theta \leq \pi$, determine the value of $\sin \theta$? 2
- g. Evaluate $\int_0^2 \frac{6x}{x^2 + 2} dx$, leaving your answer in simplest exact form. 3

End of Question 11

Question 12 Start A New Booklet**15 Marks**

- a. The line l connects the points A (2,6) and B (4,9).
Point C has the coordinates (5,4).



- | | | |
|------|--|----------|
| i. | Determine the gradient of Line l . | 1 |
| ii. | Determine the exact length AB | 1 |
| iii. | Determine the area of triangle ABC | 3 |
| iv. | What angle, to the nearest degree, does the line through AC make with the positive x – axis? | 2 |
- b. Given the simultaneous equations:

$$\begin{aligned} 2x + y &= 1 \\ x^2 - 4ky + 5k &= 0 \end{aligned}$$

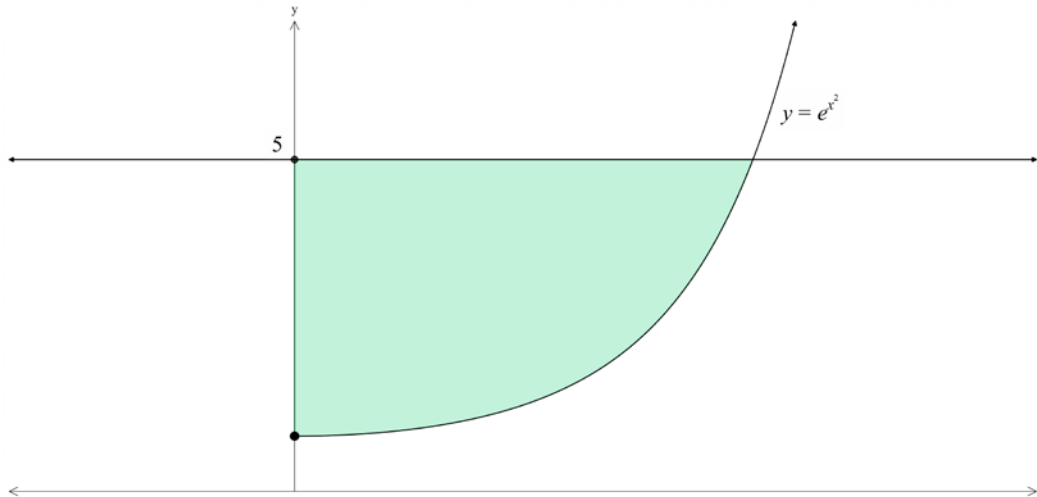
where k is a non-zero constant

- | | | |
|------|--|----------|
| i. | Show that $x^2 + 8kx + k = 0$ | 1 |
| ii. | Given that $x^2 + 8kx + k = 0$ has equal roots, find the value of k . | 2 |
| iii. | For this value of k , find the solution of the simultaneous equations. | 1 |

Question 12 continues on the next page.

Question 12 continued:

- c. The shaded region bounded by the graph $y = e^{x^2}$, the line $y = 5$ and the y – axis is rotated about the y – axis to form a solid of revolution.



- i. Show that the volume of the solid is given by

$$V = \pi \int_1^5 \ln y \, dy$$

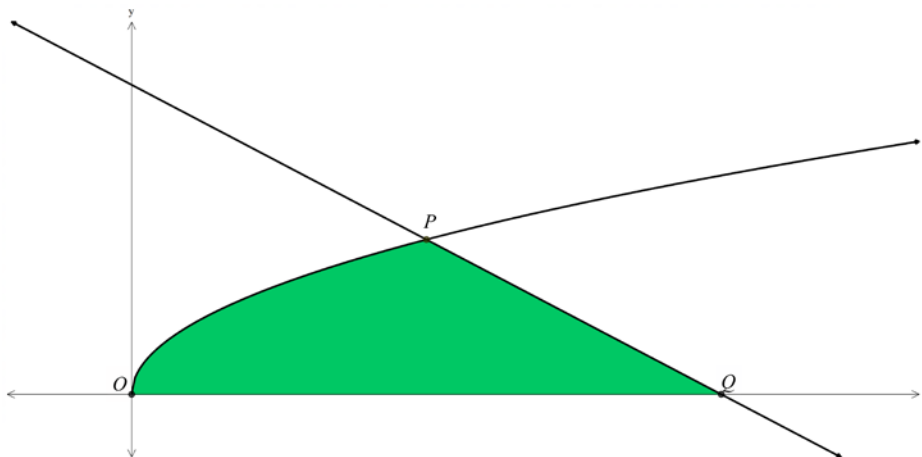
1

- ii. Use Simpsons Rule with five functions values to approximate the volume of the solid of revolution V_y correct to three decimal places.

3**End of Question 12**

Question 13 Start A New Booklet**15 Marks**

- a. The derivative of a function is given by $f'(x) = 3x^2 + 3$. The curve passes through the point $(1, 1)$. Find the equation of the curve. 2
- b. i. Find the sum of the sequence 100, 101, 102,, 999. 1
- ii. Hence, or otherwise, find the sum of all the 3 digit numbers which are not divisible by 5. 2
- c. In the diagram below, the shaded region is bound by graphs of $y = 8 - x$, $y = 2\sqrt{x}$ and the x - axis.

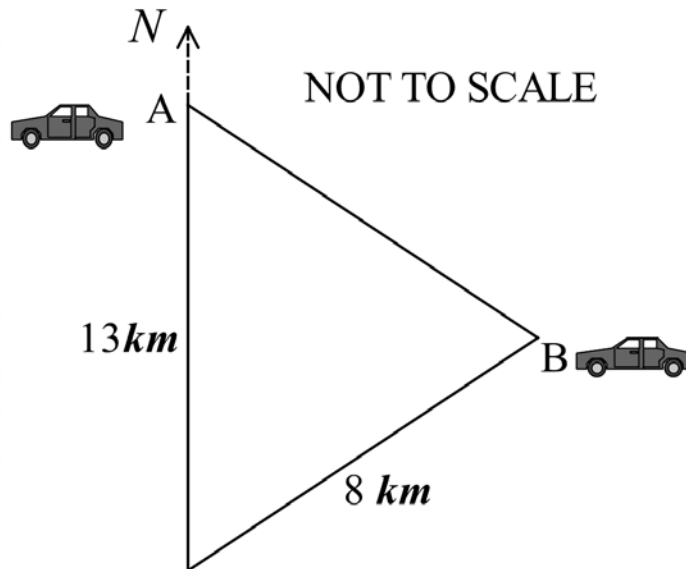


- i. Determine the coordinates of point P. 2
- ii. Find the exact value of the shaded region OQP. 2
- d. Find the equation of normal to the curve $y = \sqrt{x^2 + 7}$ at $x = 3$. 3

Question 13 continues on the next page.

Question 13 continued:

- e. Two vehicles, Car A and Car B, depart from the same starting position. Car A travels north for a distance of 13 km. At the same time Car B travels on a bearing 040° T for a distance of 8 km.

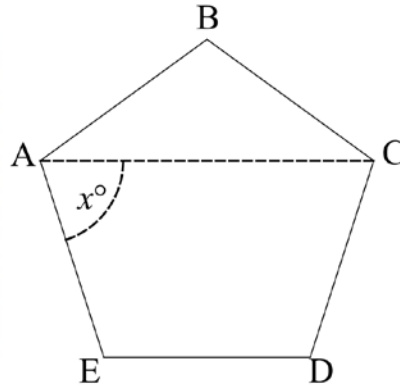


- | | | |
|-----|---|----------|
| i. | What is the distance between the cars at this time? | 1 |
| ii. | What is the bearing of Car A from Car B? | 2 |

End of Question 13

Question 14 Start A New Booklet**15 Marks**

- a. $ABCDE$ is a regular pentagon and $\angle CAE = x^\circ$



Determine the value of x , giving reasons for your answer.

2

- b. i. Differentiate $x \cos 2x$.

2

- ii. Hence find $\int_0^{\frac{\pi}{6}} x \sin 2x \, dx$

2

- c. i. Sketch the curve of $y = |x - 2| - 5$

2

- ii Hence solve $|x - 2| - 5 < 2x$

1

- d. For the curve given by $f(x) = 2x^2 e^{-x}$

- i. Find any stationary points and determine their nature

3

- ii Determine $\lim_{x \rightarrow \infty} f(x)$

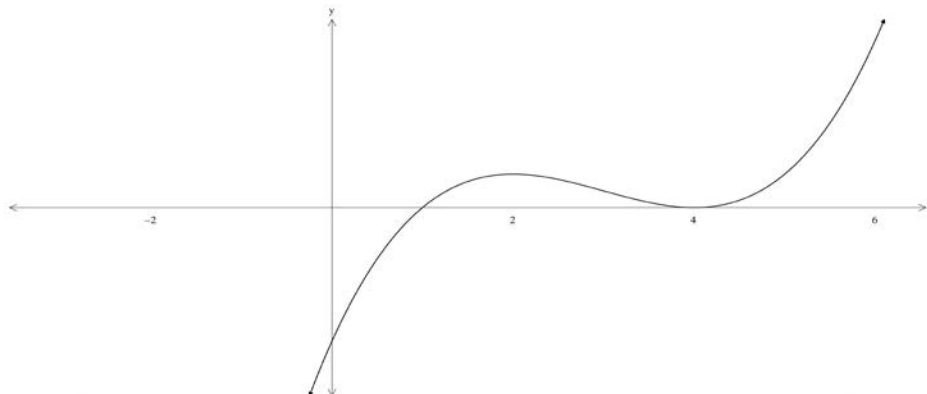
1

- iii Hence sketch the curve

2**End of Question 14**

Question 15 Start A New Booklet**15 Marks**

- a. The graph of $y = f'(x)$ is given below.



Copy or trace the graph onto your answer booklet, using at least a third of the page.

Sketch the graph of $y = f(x)$ given that it passes through the points $(0,0)$ and $(4,-2)$.

Show clearly any turning points and/or points of inflexion.

3

- b. The velocity of a particle is given by:

$$\dot{x} = 1 - 2\sin\pi t, \text{ where } x \text{ is in metres and } t, \text{ is the time in seconds.}$$

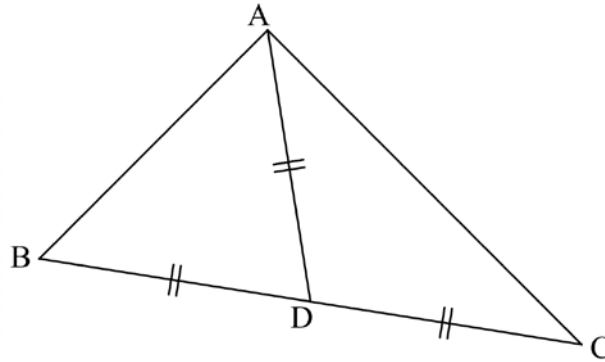
Initially the particle is $\frac{2}{\pi}$ m to the right of the origin.

- i. Find the acceleration of the particle in terms of t . **1**
- ii Find the values of t in the interval $0 \leq t \leq 4$ where the particle is at rest. **2**
- iii. Sketch the graph $\dot{x} = 1 - 2\sin\pi t$ as a function of time for $0 \leq t \leq 4$,
Show all intercepts on the horizontal and vertical axes. **3**
- ii. Find the distance travelled in the interval $t = 1$ to $t = 2$. **2**

Question 15 continues on next page.

Question 15 continued:

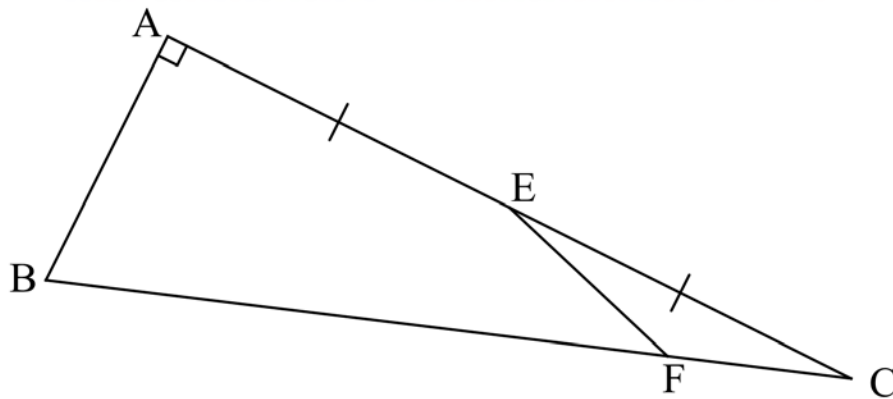
- c. In the triangle ABC , $AD = BD = CD$



- i. Prove $\angle BAC = 90^\circ$.

2

- ii. In triangle ABC , $\angle BAC = 90^\circ$, $AE = CE$, $EF = 8$ and $\frac{BC}{FC} = 4$



Using part i , or otherwise, determine the length of BC .

2

End of Question 15

Question 16 Start A New Booklet**15 Marks**

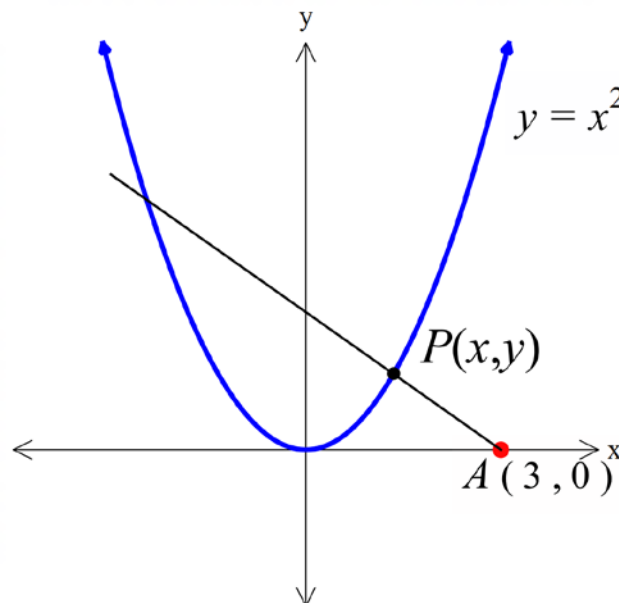
- a. A bacteria population, P , in a container is modelled by:

$$P(t) = 220 - Ae^{kt},$$

where t is in days and P expressed in thousands.
Initially there were 100 000 bacteria in the container.

- i. State the value of A . 1
- ii. If the population reaches 180 000 in two days, find the exact value of k . 2
- iii. Find the limiting bacteria population in the container. 1

- b. The diagram shows the curve $y = x^2$ and the point $P(x, y)$ and $A(3, 0)$.



- i. Show that $2(x - 3) + 4x^3 = 2(x - 1)(2x^2 + 2x + 3)$ 1
- ii. Hence determine the coordinates of P such that the distance PA is a minimum. 4

Question 16 continues on next page.

Question 16 continued:

- c. Denise and Damien borrow \$500 000 to start a business. The interest rate is 7.2% p.a. compounding monthly. They plan to repay the loan in 20 years with equal regular monthly repayments.

The amount they owe after the n -th repayment is given by:

$$A_n = Pr^n - M(1 + r + \dots + r^{n-1})$$

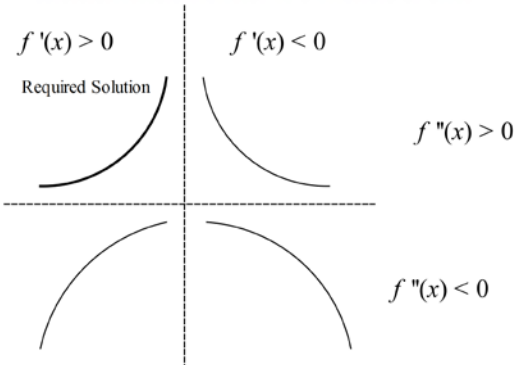
$$\text{where } P = \$500\,000, \quad r = 1 + \frac{0.072}{12}$$

and M is the amount of the regular monthly payment.

- i. Show the amount of their regular payment is \$3936.75. 2
- ii. After 4 years of regular payments, Denise and Damien make an extra payment of \$70 000. They then continue to make quarterly (once every three months) payments of \$10 000. All other conditions remain the same.

Explain if they will pay off the loan earlier than they initially planned.
Justify your answer with calculations. 4

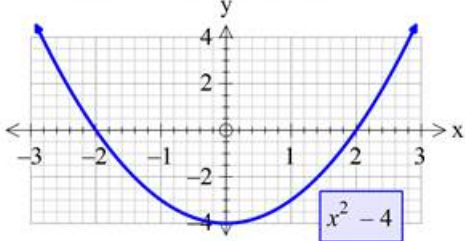
End of Examination

Multiple Choice		
Question	Answer	Solution
1	A	$x = \frac{x_1 + x_2}{2} \quad y = \frac{y_1 + y_2}{2}$ $-2 = \frac{2+x}{2} \quad 1 = \frac{-3+y}{2}$ $-4 = 2+x \quad 2 = -3+y$ $x = -6 \quad \text{and} \quad y = 5$ $\therefore A$
2	C	2 cases positive and negative $ 2x+3 \geq 7 \quad -(2x+3) \geq 7$ $2x+3 \geq 7 \quad -2x-3 \geq 7$ $2x \geq 4 \quad -2x \geq 10$ $x \geq 2 \quad x \leq -5$ $\therefore x \leq -5, x \geq 2$ $\therefore C$
3	A	$(x+2)^2 = 4y+4$ $(x+2)^2 = 4(y+1)$
4	D	$A = \frac{a}{2}(4+4-2a)$ $= 4a - a^2$ $\therefore 4 = 4a - a^2$ $(a-2)^2 = 0$ $a = 2$
5	D	
6	D	$V = \pi \int_a^b y^2 dx$ $y^2 = (e^x)^2 = e^{2x}$ $V = \pi \int_{-2}^2 e^{2x} dx$
7	B	Period: $\frac{2\pi}{k} = 3$ $k = \frac{2\pi}{3}$

8	A	$2^1 x^1 - 2^2 x^3 + 2^3 x^5 - 2^4 x^7 + \dots$<table><tr><td>Power of x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td><td>...</td></tr><tr><td>Coefficient</td><td>2^1</td><td>-2^2</td><td>2^3</td><td>-2^4</td><td>2^5</td><td>...</td></tr></table> Every 2nd term is negative, so the coefficient of x^{19} is negative. The power of x in each term is twice the power of 2, minus 1. So, if the power of x is 19, then the power of 2 is 10. $\therefore -2^{10} x^{19}$ Alternatively. Geometric Series $a = 2x \quad r = -2x^2$ $T_n = ar^{n-1} = 2x(-2x^2)^{n-1}$$= 2(-2)^{n-1} x \times x^{2n-2}$$= 2(-2)^{n-1} x^{2n-1}$ but each term in options is x^{19} therefore $n = 10$ $T_n = 2 \times (-2)^9 x^{19} = -2^{10} x^{19}$	Power of x	1	3	5	7	9	...	Coefficient	2^1	-2^2	2^3	-2^4	2^5	...
Power of x	1	3	5	7	9	...										
Coefficient	2^1	-2^2	2^3	-2^4	2^5	...										
9	B	$-2 = 2^a + b \quad (1)$ $0 = 2^{a+1} + b \quad (2)$ $0 = 2 \cdot 2^a + b$ $b = -2 \cdot 2^a$ $-2 = -2 \cdot 2^a + 2^a$ $2^a = 2$ $a = 1$ $b = -4$														
10	D	$\int_{-\frac{\pi}{2}}^k \cos x \, dx = 3 \int_k^{\frac{\pi}{2}} \cos x \, dx$ $\sin k - \sin\left(-\frac{\pi}{2}\right) = 3\left(\sin\left(\frac{\pi}{2}\right) - \sin k\right)$ $\sin k + 1 = 3 - 3\sin k$ $4 \sin k = 2$ $\sin k = \frac{1}{2}$ $k = \frac{\pi}{6}$														

Q11

a	$\frac{12x^2 - 4x - 1}{(6x + 1)(2x - 1)}$	1 mark correct solution
b	$\frac{18}{3 - \sqrt{3}}$ $= \frac{18}{3 - \sqrt{3}} \times \frac{3 + \sqrt{3}}{3 + \sqrt{3}}$ $= 18 \times \frac{3 + \sqrt{3}}{9 - 3}$ $= 18 \times \frac{3 + \sqrt{3}}{6}$ $= 3(3 + \sqrt{3})$ $= 9 + 3\sqrt{3}$ $= 9 + \sqrt{27}$ $\therefore a = 9 \text{ and } b = 27$	2 marks correct solution 1 mark correct multiplication and expansion of denominator
c	$\frac{d}{dx} \frac{2x + 1}{\sqrt{x}}$ $= \frac{d}{dx} \frac{2x + 1}{x^{\frac{1}{2}}}$ $= \frac{d}{dx} 2x^{\frac{1}{2}} + x^{-\frac{1}{2}}$ $= x^{-\frac{1}{2}} - \frac{1}{2}x^{-\frac{3}{2}}$ Product rule and quotient rule also work but are much harder to apply	2 marks correct solution 1 mark for correct simplification of both parts prior to differentiation or if used product or quotient rule correct application of formula
d	$\int_0^2 (5x - 1)^9$ $= \left[\frac{5x - 1}{10 \times 5} \right]_0^2$ $= \frac{1}{50} ((5 \times 2 - 1)^{10} - (-1)^{10})$ $= \frac{9^{10} - 1}{50}$ $= 69\,735\,688$	3 marks correct solution 2 marks correct integration and correct substitution 1 mark correct integration but incorrect or no substitution. Answer in index form preferred

e	 $f(x) = \frac{x}{\sqrt{x^2 - 4}}$ Domain will exist if $x^2 - 4 > 0$ Only part of the curve that is above but not equal to points on x axis $\therefore$ From graph above $x < -2$ and $x > 2$	2 marks correct solution 1 mark if only gives one part of solution or of gives $= -2$ or 2 as part of solution
f	$\cos(2\theta) = \frac{1}{2} \text{ for } 0 \leq \theta \leq \pi$ $\beta = 2\theta \text{ for } 0 \leq \beta \leq 2\pi$ $\cos(\beta) = \frac{1}{2}$ Using <i>ASTC</i> $\therefore$ Quad 1 and 4 $\beta = \frac{\pi}{3} \text{ and } \frac{5\pi}{3}$ $\therefore \theta = \frac{\pi}{6} \text{ and } \frac{5\pi}{6}$ $\therefore \sin\theta = \frac{1}{2}$	2 marks correct solution 1 mark if only finds angle with no value for sin
g	$\int_0^2 \left(\frac{6x}{x^2 + 2} \right) dx$ note: numerator is 3 times diff of denominator so log function use $\log_e = \ln$ $= 3 \times \int_0^2 \left(\frac{2x}{x^2 + 2} \right) dx$ $= 3 \times [\ln(x^2 + 2)]_0^2$ $= 3[\ln(6) - \ln(2)]$ $= 3\ln \frac{6}{2}$ $= 3\ln(3) \text{ or } \ln(27)$	3 marks correct solution 2 marks correct integration and substitution but not in simplest form 1 mark correct integration

Q12

ai	$m = \frac{9-6}{4-2}$ $= \frac{3}{2}$	1 mark correct solution
a ii	$d = \sqrt{(4-2)^2 + (9-6)^2}$ $= \sqrt{13}$	1 mark correct solution
a iii	$y - 6 = \frac{3}{2}(x - 2)$ $2y - 12 = 3x - 6$ $3x - 2y + 6 = 0$ $d \perp = \frac{ 3(5) - (-2)(4) + 6 }{\sqrt{(3)^2 + (-2)^2}}$ $= \frac{13}{\sqrt{13}}$ $A = \frac{1}{2}(\sqrt{13})(\sqrt{13})$ $= \frac{13}{2} u^2$	3 marks correct solution 2 mark correct solution without proof for perp. 1 mark one correct step only either in perp distance, or gradient of AC or distance AC
a iv	$m_{AC} = \frac{4-6}{5-2}$ $= -\frac{2}{3}$ $\frac{2}{3} = \tan(180 - \theta)$ $\theta = 180 - \tan^{-1}\left(\frac{2}{3}\right)$ $= 180 - 34$ $= 146^\circ$	2 mark correct solution 1 mark correct acute angle
bi	$y = 1 - 2x$ $x^2 - 4k(1 - 2x) + 5k = 0$ $x^2 - 4k + 8kx + 5k = 0$ $x^2 + 8kx + k = 0$	1 mark correct solution
b ii	$\Delta = 0$ $\therefore (8k)^2 - 4k = 0$ $64k^2 - 4k = 0$ $k = 0 \text{ or } k = \frac{1}{16}$ $\text{but } k \neq 0 \therefore k = \frac{1}{16}$	2 mark correct solution 1 mark correct solution but no rejection for k=0

biii	$x^2 + \frac{1}{2}x + \frac{1}{16} = 0$ $\left(x + \frac{1}{4}\right)^2 = 0$ $x = -\frac{1}{4}$						1 mark correct solution												
ci	$y = e^{x^2}$ $x^2 = \ln y \text{ and } x = 0, y = 1$ $\therefore V = \pi \int_1^5 x^2 dy$ $= \int_1^5 \ln y dy$						1 mark correct solution												
cii	<table><tr><td>y</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>ln y</td><td>ln 1</td><td>ln 2</td><td>ln 3</td><td>ln 4</td><td>ln 5</td></tr></table>	y	1	2	3	4	5	ln y	ln 1	ln 2	ln 3	ln 4	ln 5						3 marks correct solution
	y	1	2	3	4	5													
ln y	ln 1	ln 2	ln 3	ln 4	ln 5														
	$V = \frac{\pi}{6}[(\ln 1 + \ln 5) + 4(\ln 2 + \ln 4) + 2(\ln 3)]$ $= 12.1244$ $= 12.124 u^3$						2 mark partial correct solution with one mistake only 1 mark correct values in table												

COMMENTS

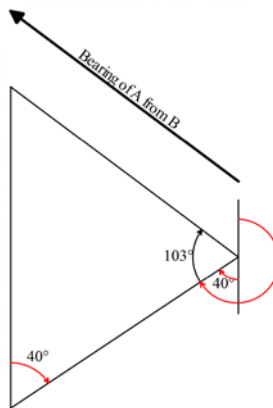
- a) iii) many students found area without first showing right angled triangle
 iv) common mistake to state the acute angle despite diagram showing an obtuse angle
- b) ii) divisions which resulted in the loss of the zero solution without reasons did not get full marks
- c) i) lower bound needed to be shown by calculation as well as manipulation of expression
 ii) common mistake in omitting pi from calculations

a	$f'(x) = 3x^2 + 3$ $f(x) = \int 3x^2 + 3 \, dx$ $= x^3 + 3x + c$ $x = 1 \Rightarrow y = 1$ $1 = 1 + 3 + c$ $c = -3$ $f(x) = x^3 + 3x - 3$	2 marks – correct solution 1 mark – correct integral.
b-i	$T_n = a + (n - 1)d$ $999 = 100 + (n - 1) \times 1$ $n = 999 - 100 + 1 = 900$ $S_n = \frac{n}{2}(a + l)$ $= \frac{900}{2}(100 + 999)$ $= 494500$	1 mark – correct answer
b-ii	Sum of multiples of 5 from 100 to 995 $T_n = 100 + (n - 1) \times 5$ $995 = 100 + 5n - 5$ $n = \frac{900}{5} = 180$ $S_n = \frac{n}{2}(a + l)$ $= \frac{180}{2}(100 + 995)$ $= 98550$ $\therefore$ nonfactors of 5 $S_n = 494550 - 98550$ $= 396000$	2 marks – correct answer 1 mark – correct n - Correct answer from incorrect n

c-i	$8 - x = 2\sqrt{x}$ let $m = \sqrt{x}$ $8 - m^2 = 2m$ $m^2 + 2m - 8 = 0$ $(m + 4)(m - 2) = 0$ $m = -4 \text{ or } m = 2$ $\sqrt{x} \neq -4$ $\therefore \sqrt{x} = 2$ $x = 4$ Coordinate $y = 8 - 4$ $P(4,4)$	2 marks – correct answer including explanation for why coordinate (4,4) was correct. 1 mark – $x = 4$ without further interpretation and/ or y - value
c-ii	Shaded Region $A = \int_0^4 2\sqrt{x} \, dx + \text{Area of Triangle}$ $= 2 \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^4 + \frac{1}{2} \times 4 \times 4$ $= \frac{4}{3} \{(\sqrt{4})^3 - 0\} + 8$ $= \frac{32}{3} + 8 = \frac{56}{3}$ Note: no marks awarded for $\int_0^8 2\sqrt{x} - (8 - x) \, dx$ or similar as this is not the defined shaded region.	2 marks – correct answer 1 mark - Either subregion correct using correct expression for total area

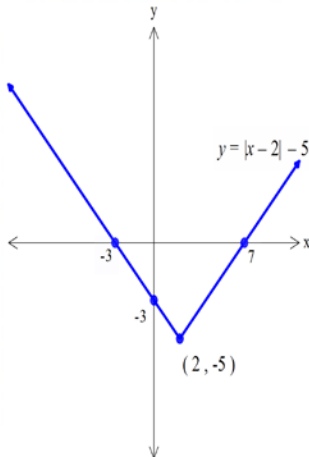
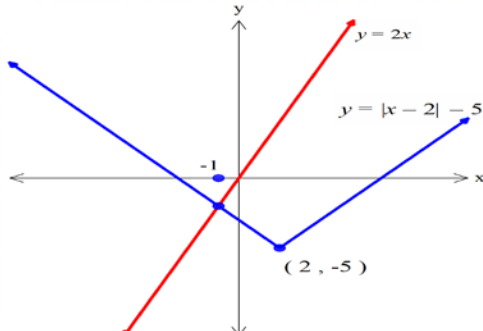
d	$y = \sqrt{x^2 + 7}$ $x = 3 \Rightarrow y = \sqrt{9 + 7} = 4$ $(3,4)$ $\frac{dy}{dx} = 2x \times \frac{1}{2} \times (x^2 + 7)^{-\frac{1}{2}}$ $= \frac{x}{\sqrt{x^2 + 7}}$ at $x = 3$ $m_{\text{Tangent}} = \frac{3}{\sqrt{16}} = \frac{3}{4}$ $\therefore m_{\text{Normal}} = -\frac{4}{3}$ $y - 4 = -\frac{4}{3}(x - 3)$ $3y - 12 = -4x + 12$ $4x + 3y - 24 = 0$	3 marks correct solution 2 marks - Correct equation from incorrect gradient - Correct gradient for the normal. 1 mark Identifying - $m_T \times m_N = -1$ - correct initial differentiation
e-i	$c^2 = a^2 + b^2 - 2ab\cos C$ $c = \sqrt{13^2 + 8^2 - 2 \times 13 \times 8 \times \cos 40}$ $= 8.5827$ $\cong 8.6\text{km}$	1 mark – correct solution .
e-ii	$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$ $B = \cos^{-1} \left(\frac{8^2 + 8.582^2 - 13^2}{2 \times 8 \times 8.582} \right)$ $= 103^\circ 11'$ If using Sine Rule – take care – 2 possible answers $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin(180 - B)}{b}$ $\frac{\sin 40}{8.5827} = \frac{\sin B}{13}$ $B = \sin^{-1} \left(\frac{13 \times \sin 40}{8.5827} \right)$ $= 76^\circ 48' \text{ or } 103^\circ 12'$ But 76° does not fit with required dimensions.	2 mark – correct solution 1 mark - correct bearing from an incorrect use of Sine rule. - Correct angle but incorrect bearing

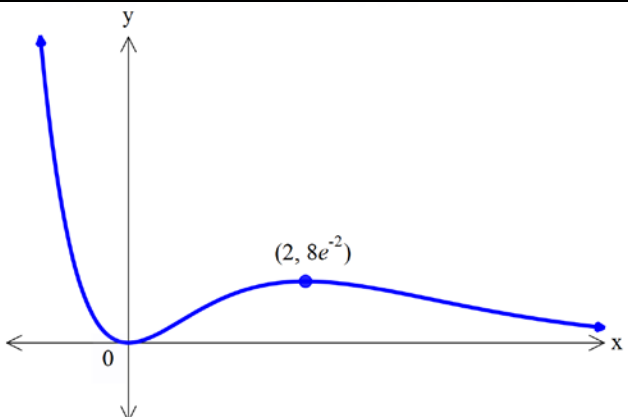
Bearing of A from B
= $103 + 40 + 180$
= 323°



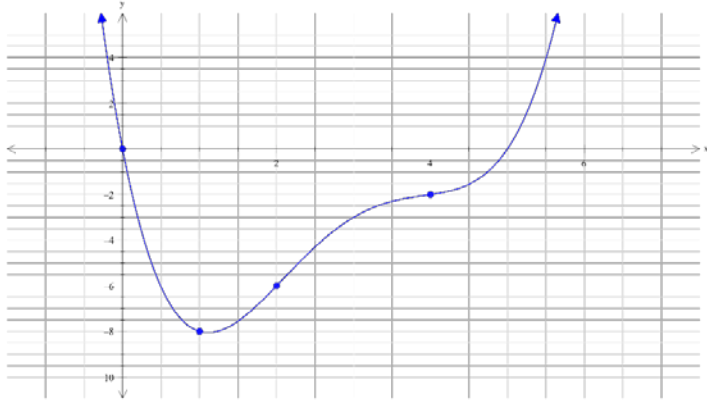
Q14

a	$\angle BAE = \frac{180(5-2)}{5} \text{ (angle sum of regular pentagon)}$ $= 108^\circ$ Similarly, $\angle B = 108^\circ$ $AB = BC$ (regular pentagon) $\therefore \triangle ABC$ is isosceles $\angle BAC = \frac{180 - 108}{2}$ (base $\angle$s of isosceles $\triangle$; angle sum of a $\triangle$) $= 36$ $\therefore x = 108 - 36$ $= 72$	2 marks 1 mark deducted for unclear/insufficient reasoning/ just the correct value of x
bi	Let $u = x$ $v = \cos 2x$ $u' = 1$ $v' = -2\sin 2x$ $\frac{d}{dx}(x \cos 2x)$ $= uv' + vu'$ $= -2x \sin 2x + \cos 2x$	2 marks 1 mark for one error
bii	From (i): $\frac{d}{dx}(x \cos 2x) = -2x \sin 2x + \cos 2x$ $\therefore -2x \sin 2x = \frac{d}{dx}(x \cos 2x) - \cos 2x$ Now, $\int_0^{\frac{\pi}{6}} x \sin 2x \, dx$ $= -\frac{1}{2} \int_0^{\frac{\pi}{6}} -2x \sin 2x \, dx$ $= -\frac{1}{2} \int_0^{\frac{\pi}{6}} \left(\frac{d}{dx}(x \cos 2x) - \cos 2x \right) dx$ $= -\frac{1}{2} \left[x \cos 2x \right]_0^{\frac{\pi}{6}} + \frac{1}{2} \int_0^{\frac{\pi}{6}} \cos 2x \, dx$ $= -\frac{1}{2} \left[x \cos 2x \right]_0^{\frac{\pi}{6}} + \frac{1}{2} \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{6}}$ $= -\frac{1}{2} \left(\frac{\pi}{6} \cos \frac{\pi}{3} \right) + \frac{1}{4} \sin \frac{\pi}{3}$ $= -\frac{\pi}{24} + \frac{\sqrt{3}}{8}$	2 marks 1 mark for a correct indefinite integral/ a correct expression using part (i) Note: Many students made errors with minus signs and fractions. Show all steps to avoid silly mistakes.
ci	Absolute value graph $y = x $ shifted right 2 units, down 5 units	2 marks

		1 mark for a different absolute value graph with correct x-intercepts Note: two points should be labelled on each branch (so that each branch is uniquely defined)														
cii	Solve $x - 2 - 5 < 2x$ Sketch $y = \text{LHS}$ and $y = \text{RHS}$ on the same set of axes, and see where the absolute value graph is underneath the line $y = 2x$  The left branch of the absolute value graph has equation: $y = -(x - 2) - 5$ $y = -x + 2 - 5$ $y = -x - 3$ So for the point of intersection: $2x = -x - 3$ $3x = -3$ $x = -1$ $\therefore x > -1$ is the solution	1 mark correct answer Note: many tried unsuccessfully to solve purely algebraically														
di	$f(x) = 2x^2 e^{-x}$ Let $u = 2x^2$ $v = e^{-x}$ $\therefore u' = 4x \quad v' = -e^{-x}$ $f'(x) = 4xe^{-x} - 2x^2 e^{-x}$ $f'(x) = 2x e^{-x}(2 - x)$ stationary points: $2xe^{-x}(2 - x) = 0$ $x(2 - x) = 0 \text{ (dividing by } 2e^{-x} \text{)}$ $\therefore x = 0 \text{ or } 2$ y-coordinates: $f(0) = 0$ $f(2) = 8e^{-2}$ <table border="1" data-bbox="202 2054 1125 2130"><tr><td>x</td><td>-0.1</td><td>0</td><td>0.1</td><td>1.9</td><td>2</td><td>2.1</td></tr><tr><td>f'(x)</td><td>-</td><td>0</td><td>+</td><td>+</td><td>0</td><td>-</td></tr></table>	x	-0.1	0	0.1	1.9	2	2.1	f'(x)	-	0	+	+	0	-	3 marks 2 marks for correctly finding and determining the nature of one stationary point 2 marks for finding both points but not determining their nature 1 mark for correct first derivative Note: Dividing both sides by x excludes the possibility that $x = 0$. When solving for x,
x	-0.1	0	0.1	1.9	2	2.1										
f'(x)	-	0	+	+	0	-										

	Pictures showing sign of gradient \ — / / — \ Therefore, (0,0) is a minimum and (2, 8e⁻²) is a maximum turning point.	bring all terms to one side (= 0), then factorise and use the Null Factor Law.
dii	$\lim_{x \rightarrow \infty} 2x^2 e^{-x}$ $= \lim_{x \rightarrow \infty} \frac{2x^2}{e^x}$ $= 0$	1 mark correct answer
diii		2 marks for a graph showing all the results from d(i) and (ii) 1 mark deducted for a missing feature/ an incorrect feature (e.g. an extra stationary point)

Question 15

a)	 Horizontal point of inflection at (4,-2) Minimum T.P. on line x=1 eg (1,-8) Point of inflection on line x=2 eg (2,-6)	3 marks correct solution 2 marks 2 key points labelled of Min TP POI HPOI 1 mark correct shape
bi)	$\ddot{x} = -2\pi \cos \pi t$	1 Correct answer

ii)	$\dot{x} = 1 - 2 \sin \pi t$ $\sin \pi t = \frac{1}{2}$ $\pi t = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$ $t = \frac{1}{6}, \frac{5}{6}, \frac{13}{6}, \frac{17}{6}$	2 marks correct answer 1 mark 2 t values found
iii)	P= 2 so 2 cycles range [-1,3] inverted raised 1 unit x ints from part ii)	3 marks correct answer 2 marks 1 mark correct correct y int and
iv)	$\int_1^2 1 - 2 \sin \pi t \, dt = \left[t + \frac{2 \cos \pi t}{\pi} \right]_1^2$ $= 2 + \frac{2 \cos 2\pi}{\pi} - \left(1 + \frac{2 \cos \pi}{\pi} \right)$ $= 2 + \frac{2}{\pi} - \left(1 - \frac{2}{\pi} \right)$ $= 1 + \frac{4}{\pi} \text{ or } \frac{4 + \pi}{\pi}$	2 marks correct solution 1 mark correct integration
ci)	In $\triangle ABD$ Let $\angle DBA = \alpha$. Then $\angle BAD = \alpha$ (isosceles $\triangle$) In $\triangle ABD$ Let $\angle DAC = \beta$. Then $\angle DCA = \beta$ (isosceles $\triangle$) Angle sum of $\triangle ABC = 2\alpha + 2\beta = 180^\circ$ $\alpha + \beta = 90^\circ = \angle BAC$ As required	2 marks correct solution 1 some progress
ii)	Let $CF = x$, then $BC = 4x$ Draw $EM \perp AB$. From part (i): $EF = FM = CF = 8 = x \Rightarrow BC = 4x = 32$.	2 marks correct solution 1 mark $CF=EF$ with reasons

Comments: Sketches using scale should be smooth curves and actually resemble the points featured.
b) SIN Sine Integrates Negative – double check with negatives c) proof format was mostly NOT adopted

Question 16

(a)(i)	$P(0) = 100 \Rightarrow A = 120$	1 mark: correct answer								
(a)(ii)	$220 - 120e^{2k} = 180 \Rightarrow e^{2k} = \frac{1}{3} \Rightarrow k = \frac{1}{2} \log_e \frac{1}{3}$	2 marks: correct answer 1 mark: progress towards solution								
(a)(iii)	As $t \rightarrow \infty$, $-120e^{kt} \rightarrow 0$, $P(t) \rightarrow 220$ $\therefore 220\,000$	1 mark: correct answer (no mark for 220)								
(b)(i)	$\begin{aligned} \text{RHS} &= (2x - 2)(2x^2 + 2x + 3) \\ &= 4x^3 + 4x^2 + 6x - 4x^2 - 4x - 6 \\ &= 2x - 6 + 4x^3 \\ &= 2(x - 3) + 4x^3 \\ &= \text{LHS} \end{aligned}$	1 mark: correct answer								
	Marker's comments: Incorrect to treat as equation and solve or substituting an x-value to show both sides equal. Mark was awarded for subtracting expressions and finding equal to zero but this is not the normal way to show equivalence in expressions.									
(b)(ii)	$\begin{aligned} \text{PA}^2 &= (x - 3)^2 + y^2 = (x - 3)^2 + x^4 \\ \text{let } s &= (x - 3)^2 + x^4 \\ \frac{ds}{dx} &= 2(x - 3) + 4x^3 = 2(x - 1)(2x^2 + 2x + 3) \text{ from part (i)} \\ \text{stat pts at } \frac{ds}{dx} &= 0 \\ x - 1 &= 0 \text{ or } 2x^2 + 2x + 3 = 0 \\ x &= 1 \quad \text{No solution} \\ \text{at } x &= 1, \quad y = 1 \\ \therefore P(1,1) \end{aligned}$ Alternative method: $\begin{aligned} \text{PA} &= ((x - 3)^2 + x^4)^{\frac{1}{2}} \\ \text{PA}' &= \frac{1}{2}((x - 3)^2 + x^4)^{-\frac{1}{2}} [2(x - 3) + 4x^3] \\ &= \frac{2(x - 3) + 4x^3}{2\sqrt{(x - 3)^2 + x^4}} \\ &= \frac{2(x - 1)(2x^2 + 2x + 3)}{2\sqrt{(x - 3)^2 + x^4}} \\ \text{stat pts at } \text{PA}' &= 0 \\ \text{solution then follows as above} \end{aligned}$ <table border="1"><tr><td>x</td><td>1^-</td><td>1</td><td>1^+</td></tr><tr><td>$\frac{ds}{dx}$</td><td>neg</td><td>0</td><td>pos</td></tr></table> Therefore minimum at P(1,1)	x	1^-	1	1^+	$\frac{ds}{dx}$	neg	0	pos	4 marks: correct answer 3 marks: correct solution without showing that the point is a minimum 2 marks: correct distance equation and derivative 1 mark for correct distance equation in terms of x
x	1^-	1	1^+							
$\frac{ds}{dx}$	neg	0	pos							

(c)(i)	Number of payments = $20 \times 12 = 240$ $r = 1 + \frac{0.072}{12} = 1.006$ $(1 + r + \dots + r^{n-1}) = \frac{r^{240} - 1}{r - 1}$ (sum of geometric series with $a = 1, n = 240$) $A_{240} = 0 \Rightarrow Pr^{240} - M \frac{r^{240} - 1}{r - 1} = 0$ Substituting P and r, $M = \frac{Pr^{240}(r - 1)}{r^{240} - 1} = \\3936.75	2 marks: correct answer 1 mark: progress towards solution
(c)(ii)	Loan after extra payment (for $M = \\$3936.75$): $A_{48} - 70\,000 = \$378071.26 = Q$ Let B_n be the amount owing after the first payment of new schedule, $N = \\$10000$ and $s = r^3$. $B_n = Qs^n - N(1 + s + s^2 + \dots + s^{n-1}) = Qs^n - N \frac{s^n - 1}{s - 1}$ $B_n = 0 \Rightarrow sQs^n - Qs^n - Ns^n + N = 0$ $s^n = \frac{-N}{sQ - Q - N} \Rightarrow n = \log_s \left(\frac{-N}{sQ - Q - N} \right)$ Substituting s, Q and N, $n = 64.3$ Since payments are made quarterly, loan will be paid off in 16.08 years. This is longer than the remaining term, 16 years. Alternative methods: 1) Substituting $n=64$ (number of quarters in 16 years) into B_n to find that $B_{64} = 2983.92$ (loan amount remaining after 20 years) 2) Substituting $n=64$ and $B_n = 378071.26$ to find $N \approx 10025.09$	4 marks: correct answer 3 marks: correct number of payments 2 marks: correct formula for amount owing in new schedule 1 mark: correct loan after extra payment
Marker's comments: The question did not require showing how the formula A_n was derived. Students were able to use it directly, yet many still found A_1, A_2 etc (in both part (i) and (ii)) Common error in part (ii) was using the incorrect rate. Students who interpreted the question as the interest being compounded quarterly used $r = 1 + \frac{0.072}{4} = 1.018$ instead of $r = 1.006^3 \approx 1.018108216$		